The Open University of Sri Lanka
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Bachelor of Software Engineering Honors
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Test Plan

for

VIZHIYAL

[Event Management System]

Group – 30

Supervisor : Dr. H.M.M.Caldera

EEY6689 - Event Management System – TEST PLAN - V1.0

Team

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Table of Contents

1. Test Plan Identifier

Master Test Plan for the **Vizhiyal Event Management System.**

2. References

The following documents and standards support this Test Plan:

- Project Plan
- Project Proposal
- System Requirement Specification (SRS)
- Presentation
- User Interface Design Document

3. Introduction

Objective of the Plan

The objective of this Test Plan is to define the testing strategy, scope, objectives, schedule, and resources required to verify and validate the functionality, reliability, and performance of the **Vizhiyal Event Management System.** This plan ensures that all modules of the system meet specified requirements and perform as expected before deployment.

Scope of the Plan

This Test Plan covers functional, integration, system, and user acceptance testing for the **Vizhiyal Event Management System.** The testing will focus on modules such as user registration and login, event browsing, vendor management, booking system, payment gateway, and progress tracking. Non-functional testing such as usability and performance testing will also be conducted where applicable.

Relation to the Software Project Plan

This plan is aligned with the overall **Vizhiyal Project Management Plan.** It follows the same timelines, milestones, and quality objectives defined in the project lifecycle. The testing outcomes will directly contribute to achieving the project's quality assurance goals.

Constraints

Testing will be conducted within the allocated project timeline and resource limitations. The testing team will use the available development environment, with a defined budget for tools and test data preparation.

Coordination and Change Control

All change requests and test-related updates will be communicated through the Configuration Management process. Coordination among development, design, and testing teams will be managed via scheduled review meetings and documentation updates in the shared repository.

4. Test Items (Functions)

The following components and functions of the **Vizhiyal Event Management System** will be tested to ensure that the application performs according to the specified requirements and delivers the intended functionality to end users.

- **User-Related Modules**
  - **User Registration & Login:** Verify user account creation, authentication, and password recovery.
  - **User Dashboard:** Validate personalized dashboard view, booking history, and progress tracking.
  - **Profile Management:** Ensure users can update personal details, view preferences, and change passwords.
- **Vendor-Related Modules**
  - **Vendor Registration & Verification:** Confirm that vendors can register, upload required documents, and receive admin approval.
  - **Vendor Dashboard:** Test functionality for managing services, packages, and recent projects.
  - **Service Listing & Availability:** Verify vendor ability to display services, manage availability calendar, and update pricing.
- **Event Management**
  - **Event Creation & Booking:** Test user ability to select event type (e.g., wedding, birthday, corporate event), book vendors, and confirm event details.
  - **Package Customization:** Verify that users can customize vendor packages (e.g., decoration, photography).
  - **Progress Tracking:** Confirm that event status updates and milestones are visible to both users and vendors.
- **Search & Filtering**
  - **Vendor Search & Filter Page:** Test the functionality for filtering vendors by category, location, price, and rating.

- **AI-Based Suggestions:** Verify that the system recommends suitable vendors based on event type and area.

➢ **Payment & Notification System**

- **Payment Gateway Integration:** Validate secure online payment, transaction logging, and confirmation receipts.
- **Notification & Alerts:** Ensure timely email/SMS/app notifications for booking confirmations, cancellations, and updates.

➢ **Administration Functions**

- **User and Vendor Management:** Test the admin's ability to approve, edit, or deactivate accounts.
- **Review and Feedback Monitoring:** Verify that the admin can view and manage user feedback.
- **Report Generation:** Validate generation of reports such as bookings, revenue, and system activity logs.

➢ **Technical Configurations**

- **System Compatibility:** Ensure smooth functionality across browsers and devices.
- **Version Control:** Testing applies to the current stable version — *Vizhiyal v1.0* under the local CM repository.

5. Software Risk Issues

The following potential risks have been identified for the **Vizhiyal Event Management System** project. These risks may affect software quality, delivery schedule, and overall system performance if not managed properly.

- **Technical Risks**

1. **Third-Party Integrations:**
    - Risk in integrating secure payment gateways and external messaging APIs (SMS/Email services).
    - Delay or failure in third-party API updates could affect system functionality.
2. **New Technology Stack Usage:**
    - The project employs modern frameworks (e.g., React/Next.js, Node.js) that may be new to some developers.
    - There is a learning curve for effective usage, deployment, and debugging.
3. **Complex Functionalities:**

- The AI-based vendor recommendation and calendar-based availability features involve complex logic and data handling.
- Improper implementation may lead to inaccurate recommendations or booking conflicts.

4. **Interfacing Software Dependencies:**
   - Integration between user, vendor, and admin modules must remain synchronized; version mismatches could lead to data inconsistency.
5. **Data Security & Privacy Risks:**
   - Since the system handles sensitive user and vendor data, improper encryption or authorization mechanisms may expose information.

- **Project and Documentation Risks**

1. **Incomplete or Ambiguous Requirements:**
   - Misunderstanding of functional requirements between users, developers, and testers may cause incorrect implementations.
2. **Poor Documentation or Change Requests:**
   - Late-stage requirement changes or insufficient documentation of modules can cause rework and delays.
3. **Version Control Errors:**
   - Inadequate CM practices could result in overwriting or loss of updated components.

- **Operational and External Risks**

1. **Client Impact:**
   - Downtime or incorrect vendor listings can negatively affect user trust and vendor relationships.
2. **Compliance and Legal Risks:**
   - The system must comply with government data protection laws and payment processing standards (e.g., GDPR, PCI-DSS).
3. **Server or Network Failures:**
   - Hosting or database server issues could disrupt system availability.

- **Historical and Testing-Related Risks**
  1. **Defect Clustering:**
     - Early unit testing has shown that integration points (search filters, booking confirmation) are more error-prone. These areas require focused regression testing.
  2. **Insufficient Test Coverage:**
     - Limited test data or missing test cases may result in undetected bugs in rarely used modules.
- **Risk Mitigation Strategies**
  - Schedule technical reviews for high-risk modules.
  - Use automated test suites for regression and integration testing.
  - Conduct early API and interface validation.
  - Maintain proper versioning and backup of each release.
  - Perform regular audits on data handling and access control.

6. Features to Be Tested

The following features of the **Vizhiyal Event Management System** will be tested to ensure that they meet the user's expectations and functional requirements. Each feature is evaluated based on its impact and risk level to the user experience.

Feature	User Description (from User's Viewpoint)	Risk Level (H/M/L)
User Registration & Login	Users should be able to easily register using email/phone, log in securely, and manage their profile without technical issues.	M
Event Browsing & Search	Users can search for events (weddings, birthdays, corporate functions, etc.) using filters such as location, date, and category.	H
Vendor Listing & Details	Users can view vendor profiles, services offered, packages, and ratings in a well-organized format.	M
Vendor Booking System	Users can book vendors for selected dates, confirm services, and receive confirmation notifications.	H

Payment Gateway Integration	Users can make secure payments through online payment methods; payment confirmation should be instant and error-free.	H
AI-Based Vendor Recommendation	The system suggests suitable vendors based on user preferences, event type, and area.	H
User Dashboard	Users can view booking status, payment history, and progress tracking in a personalized dashboard.	M
Vendor Dashboard	Vendors can manage profiles, update packages, view booking requests, and track customer interactions.	M
Progress Tracking System	Users can monitor the status of booked services and communicate with vendors in real-time.	H
Admin Management	Admins can verify vendors, monitor user activities, and manage the overall platform effectively.	H
Notification System	Users and vendors receive instant notifications for confirmations, changes, or updates.	M
Feedback & Rating Module	After service completion, users can rate and review vendors.	L
Report Generation	Admins can generate reports related to bookings, revenue, and vendor performance.	M
Help & Support	Users can access FAQs, chat support, or contact help desk for assistance.	L

7. Features Not to Be Tested

The following features or functions will **not** be tested during this testing cycle of the **Vizhiyal Event Management System.** These exclusions are based on the current project scope, stability of existing modules, and release priorities.

Feature	Reason for Exclusion	Remarks
Multilingual Support (Tamil, Sinhala, etc.)	Not included in the current release; planned for future version to improve regional accessibility.	Will be tested in the next release cycle.

Email Marketing Automation	Low priority feature; not critical for core event management functionality in this release.	Deferred to later phase.
Third-Party Advertisement Integration	Feature not implemented yet; awaiting third-party approval and APIs.	To be included in Version 2.0.
Mobile App Version (Android/iOS)	Current release focuses on the web-based system; mobile version will be tested separately.	Mobile testing scheduled after web version stabilization.
Data Analytics Dashboard (Admin Insights)	Under development; incomplete data pipeline prevents full testing.	Expected in future update.
Chatbot/AI Assistant	Feature not yet stable and not part of MVP (Minimum Viable Product).	Post-launch enhancement feature.
Bulk Email/SMS Notifications	Considered low risk; previously tested in a similar system and found stable.	Minimal verification only.
Offline Access Features	Not supported in current system design; intended for future offline mode enhancement.	Out of current release scope.

8. Approach (Strategy)

The testing strategy for the **Vizhiyal Event Management System** focuses on ensuring that the system meets its functional and non-functional requirements through structured, multi-level testing. The testing approach is both **systematic and iterative**, covering the full lifecycle from unit testing to acceptance testing, ensuring quality at every stage.

8.1 Testing Levels

1. **Unit Testing:**
   Each module (user registration, vendor management, booking, etc.) will be tested individually by developers to verify internal logic and code correctness.

2. **Integration Testing:**
   Ensures that all integrated modules such as user-vendor interaction, event booking, and payment gateway communicate properly and exchange data accurately.

3. **System Testing:**
   Complete system validation will be performed to ensure that all functional and non-functional requirements are met in the deployed environment.

4. **User Acceptance Testing (UAT):**
   Conducted with end-users and vendors to validate usability, accuracy, and performance of the system before final release.

8.2 Testing Techniques

- **Black Box Testing:** To verify the system from a user's perspective without inspecting internal code.
- **White Box Testing:** Used during unit and integration testing to ensure code-level accuracy.
- **Regression Testing:** Performed after major bug fixes or new module integrations to ensure previously working features remain unaffected.
- **Exploratory Testing:** To identify unexpected issues not covered by test cases.

8.3 Special Tools

- **Selenium** – for automated browser-based testing.
- **Postman** – for API and backend communication testing.
- **JMeter** – for load and performance testing.
- **GitHub / GitLab** – for version control and change tracking.
- **TestRail / Excel Sheets** – for maintaining and tracking test cases and results.

8.4 Metrics Collected

Metric	Purpose	Collection Level
Number of Test Cases Executed	Measure progress	System & UAT
Pass/Fail Ratio	Assess quality	All levels
Defect Density	Identify risk areas	Integration & System
Mean Time to Resolve (MTTR)	Track defect resolution speed	Project Level
Test Coverage (%)	Evaluate completeness	System Level

8.5 Configuration Management

All testing artifacts (test cases, logs, scripts, and reports) will be maintained under **Configuration Management** using **GitHub**. Each test version and test result will be tagged according to the release version of the software (e.g., *Vizhiyal_V1.0_TestResults*).

8.6 Test Configurations

- **Hardware:**
  Standard PC configuration (Intel i5+, 8GB RAM, 512GB SSD).
- **Software:**
  Windows 10/11, modern browsers (Chrome, Firefox, Edge).
- **Database:**
  MySQL 8.0 or higher.
- **Server Environment:**
  Node.js (Backend), React (Frontend), Apache/Nginx Hosting Environment.

Each configuration will be tested in at least two combinations to ensure compatibility and performance stability.

8.7 Regression Testing

Regression testing will be performed after every major build. Priority will be given to defects classified as **High** or **Critical Severity**. Regression cycles will focus on previously affected modules.

8.8 Handling of Untestable or Unclear Requirements

Any requirement that lacks sufficient clarity, documentation, or is found to be untestable will be reported to the **Project Manager** and **Business Analyst** for clarification before test case preparation.

8.9 Organizational Process and Meetings

- **Weekly QA Review Meetings** to discuss test progress, defect reports, and blockers.
- **Daily Stand-ups** during testing phases to synchronize with development teams.
- **Post-Test Review Session** after each test phase to evaluate coverage and quality metrics.

This approach ensures structured testing of all functional, performance, and integration aspects of the **Vizhiyal Event Management System**. It uses a combination of manual and automated testing strategies to deliver a stable and high-quality system ready for deployment.

9. Item Pass/Fail Criteria

Each test case will have specific conditions that determine if it passes or fails.

- **Pass:** The system produces the expected result without any errors.
- **Fail:** The system behaves unexpectedly or produces incorrect results.
  All critical and major defects must be resolved before moving to the next testing phase.

10. Showstopper Issue

A showstopper issue is a **critical defect** that prevents further testing or system use.
Examples:

- Application crash
- Inability to log in
- Data loss or corruption
  Testing will be paused until the issue is resolved.

11. Resumption Criteria

Testing will **resume** only after:

- Showstopper defects are fixed and verified.
- The affected components have been retested.
- The test environment is stable again.

12. Remaining Test Tasks

If the application is released in **multiple phases**, some features or modules may not be covered in this plan.
These will be listed as *pending test tasks* for the next release.
This helps avoid confusion or unnecessary defect reports for incomplete functions.

13. Environmental Needs

The testing process may require:

- Specific hardware or simulators.
- Prepared test data sets.
- Stable power and network.
- Correct versions of supporting software.
  All required tools and environments must be ready before testing begins.

14. Staffing and Training Needs

Testers and team members must be trained in:

- Using the test tools.
- Understanding the system or application.
  Training ensures accurate and efficient testing.

15. Responsibilities

Defines **who is responsible** for each task:

- Test Lead: Oversees all testing activities and strategy.
- QA Engineer: Executes and reports test results.
- Developer: Fixes defects found during testing.
- Project Manager: Approves major decisions and manages schedules.

16. Schedule

Testing schedules are **linked to development milestones**.
If development is delayed, testing dates will shift accordingly.
This avoids blaming the testing team for project delays and ensures realistic timelines.

17. Planning Risks and Contingencies

Possible risks:

- Lack of test staff or tools.
- Late software delivery.
- Requirement changes during testing.
  Mitigation strategies:
- Adjust schedules or scope.
- Add resources.
- Increase acceptable defect limits (if agreed).

18. Approvals

Lists who can approve the completion of testing:

- Our Mentor
- Team Members

19. Glossary

A section explaining all **technical terms and acronyms** used in the document to ensure clear understanding and consistent communication among all stakeholders.

Thank

You